KSISU: Evidence-First District Skill Intelligence

A reproducible method for sparse public labour-market evidence, explicit uncertainty, and data-acquisition prioritisation.

Sector: Manufacturing **NCO scope:** 72, 74 **FY:** 2022-23 **Monte Carlo:** 5,000 shared draws

What the prototype demonstrates	What it does not claim
31 raw LGD districts; 30 analytical geographies after Ballari-Vijayanagara harmonisation	A proxy-only district ratio is a finding
Separate state demand evidence and district supply evidence	The partial NCVT subset is a district census
Shared uncertainty and reproducible runs	An official KSI formula
Human review for unmatched data	A policy call without full district supply
Value-of-information ranking	Validation of the proxy district ranking

Headline: public data can support an auditable inference chain, but the first policy output should often be a request for better district evidence.

1. Evidence ladder and data grain

Input	Grain	Year	Role
LGD district register	district	2026	canonical join key
Udyam district employment	district	2022 / 2023 publication	district allocation proxy
ASI Karnataka manufacturing	state	2022-23	state manufacturing employment base
NCVT MIS Karnataka ITI count	state	2022	state ITI seat magnitude
Historical district ITI counts	district weight	2022	allocation weights only; historical CTS geography
DGT/DITE tracer	division	2020-21	historical rate calibration
NCVT MIS File 1171 sample	district x ITI x trade	historical indexed	partial evidence only
Occupation mix	state prior	2022-23	analyst prior until PLFS microdata is loaded

Audit field	Purpose
file_id / SHA-256	stable artifact identity and tamper detection
source_id / URL / locator	reproducible source trail
period / grain	prevents silent state/district mixing
match method / confidence	shows how linkage was obtained
coverage status	controls whether policy output is eligible

The NCVT state page reports Karnataka at 1,467 ITIs and 216,788 seats available for 2022. The model uses that as the statewide seat magnitude. Historical district ITI counts are allocation weights only: **30/30** analytical weights are independently checked against Rajya Sabha Q131 Annexure-IV in this build and **0** remain unverified. The source uses the 2022 CTS geography; Ballari-Vijayanagara is harmonised analytically. Udyam employment is an all-sector MSME district proxy with uneven coverage and is not equivalent to manufacturing employment.

2. Linkage and harmonisation

District: normalized exact -> production alias -> fuzzy match at 0.90 -> human review. Unmatched rows remain in the review queue.

Industry: NIC retained at the first two digits when a source code exists; labels without a code remain reviewable.

Course/trade: versioned human-reviewed NCO crosswalk. The project labels the crosswalk as an analyst mapping aid, not an official course-to-NCO master.

Validation result	Value
LGD districts	31 / 31 raw
Analytical geographies	30
Seeded typo cases	80
Correct typo recovery	80 / 80 = 100.0%
Unseen aliases matched	0 / 20
Unseen aliases routed to review	20 / 20
Duplicate artifacts injected / removed	5 / 5
Unmatched rows dropped	0

Interpretation: unseen aliases are not counted as successful matches. The test is a review-queue test as well as a typo-recovery test.

3. Demand model - one basis, explicit priors

District manufacturing employment is allocated from the Karnataka ASI manufacturing total using Udyam employment shares. This keeps both sides on a consistent manufacturing base; it does not multiply Udyam all-sector employment by a manufacturing share.

NCO group	Share of total manufacturing	Status
72	0.25	modeled
74	0.15	modeled
Outside pilot	0.60	not modeled

District allocation uncertainty: lognormal sigma = 0.35 (analyst assumption; no empirical district calibration yet).
Annual openings: O(d,o) = D(d,o) x opening rate, with 5% to 12% range and 8.5% mode.

Demand uncertainty	Low	Mode	High
Opening rate	5.0%	8.5%	12.0%
Occupation prior multiplier	0.80x	1.00x	1.20x

The occupation mix is the weakest demand input until PLFS occupation-by-industry microdata is extracted. This parameter is therefore part of the uncertainty and sensitivity story, not hidden inside a point estimate.

4. Tumakuru vs Kalaburagi - partial NCVT evidence is a floor

The bundled File 1171 rows are indexed NCVT MIS evidence, not a downloaded complete district extract. Observed seats are treated as a floor plus an uncertain remainder. Therefore a partial file can widen the interval rather than create fake precision.

District	NCO	Observed seats	Before width	After width	Change	After P10 / P50 / P90	Policy
Tumakuru	72	80	2.268	2.317	-2.2%	0.85 / 1.68 / 3.17	NEEDS DATA
Tumakuru	74	60	1.755	1.793	-2.2%	0.66 / 1.30 / 2.45	NEEDS DATA
Kalaburagi	72	60	1.949	1.975	-1.4%	0.64 / 1.30 / 2.61	NEEDS DATA
Kalaburagi	74	n/a	1.508	1.508	0.0%	0.50 / 1.01 / 2.01	NEEDS DATA

What this page proves: the same model can ingest real district/trade rows without treating them as complete. The current subset is too small for a district-wide policy call. A complete 31-district x trade file is the next evidence gate.

Observed indexed rows	Seats
Tumakuru - Govt ITI Tumkur - Fitter	20
Tumakuru - Govt ITI Tumkur - Electrician	20
Tumakuru - Govt ITI Tumkur - Welder	40
Kalaburagi - Govt ITI Wadi - Fitter	20
Kalaburagi - Govt ITI Wadi - Welder	40

Absolute interval-width change (%) in the partial evidence test

5. Supply model and conservation check

S(d,o) = seats(d,c) x fill x completion x sector-entry x crosswalk(c->o). Seats, completions and sector entry remain separate quantities.

Input	Current handling
State ITI seats	216,788 seats available in NCVT MIS state row; state magnitude only
Selected course weight	0.82 across NCO 72 + 74; not normalized
Fill rate	50% / 81.3% / 90%
Completion rate	50% / 59.3% / 70%
Sector entry	20% / 30.0% / 40%
Allocation	NCVT district ITI-count weights when supplied; otherwise explicit equal-within-division fallback

Conservation diagnostic	Value
Expected mean statewide supply	23,512
Sum of district/group supply medians	21,988
Median/expected ratio	0.935
Decision	PASS - median within 25% of the triangular-mean expectation

The conservation check is designed to catch the former second-division bug: proxy seats are already split to districts, so the common allocation draw is a multiplier with mean 1.0 rather than another 1/n share. Historical district ITI counts are allocation weights only; they are not presented as current district seat totals.

6. Relative index and data-collection priority

SCI(d) = weighted share of modeled openings covered at R=min(S/D,1). For proxy-only data, the district ranking is explicitly a **data-collection view**, not a validated labour-market finding; ranking differences are driven in part by allocation weights and Udyam proxy coverage.

R rank	District	Median R	R / median	SCI	Openings	Data priority
3	Tumakuru	1.548	1.94x	1.000	1,325	1,259
16	Belagavi	0.781	0.98x	0.781	2,158	1,035
4	Vijayapura	1.530	1.92x	1.000	870	897
30	Bengaluru Urban	0.147	0.18x	0.147	10,341	788
15	Kolar	0.815	1.02x	0.815	1,282	687
10	Bagalkot	0.977	1.22x	0.935	1,022	662
6	Kalaburagi	1.196	1.50x	1.000	793	638
14	Ballari-Vijayanagara	0.844	1.06x	0.844	1,241	633
2	Ramanagara	1.985	2.49x	1.000	374	608
7	Bidar	1.177	1.48x	0.998	723	558
21	Hassan	0.672	0.84x	0.672	1,227	550
5	Chikkaballapur	1.271	1.59x	1.000	526	539

State median SCI = **0.798**. The ranking table is based on the **uncapped median ratio R**; SCI is reported separately as the capped coverage index. The district proxy ranking is not presented as validation because district supply still depends on proxy allocation. The correct policy use is: **which district should KSISU onboard first?**

Current proxy allocation diagnostic: correlation with Udyam demand share = **0.438**. This is reported as a diagnostic only; it is not proof that the proxy ranking is correct.

Top proxy ratio values - use for data collection, not policy claims

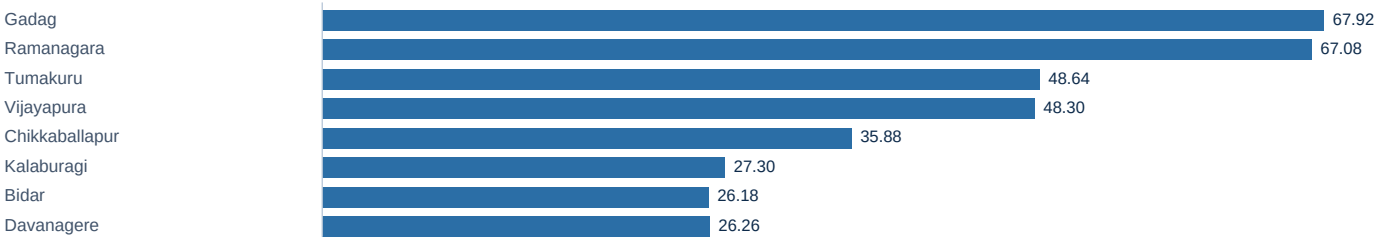
7. Sensitivity as rank stability, not range arithmetic

The model uses shared Monte Carlo draws to test whether the **uncapped ratio ordering** is stable under common uncertainty. Rank stability is a simulation diagnostic only - it is **not validation** of district labour-market reality.

District	Baseline rank	P10	Median	P90	Top-5 frequency	Global rho
Gadag	1	1	3	13	67.9%	0.740
Ramanagara	2	1	3	12	67.1%	0.740
Tumakuru	3	2	6	13	48.6%	0.740
Vijayapura	4	2	6	14	48.3%	0.740
Chikkaballapur	5	2	8	18	35.9%	0.740
Kalaburagi	6	3	9	18	27.3%	0.740
Bidar	7	3	9	18	26.2%	0.740
Davanagere	8	3	10	19	26.3%	0.740
Haveri	9	4	12	21	19.0%	0.740
Bagalkot	10	4	12	20	16.0%	0.740
Chitradurga	11	5	13	21	13.9%	0.740
Mandya	12	4	13	22	15.8%	0.740
Chamarajanagar	13	3	14	25	20.7%	0.740
Ballari-Vijayanagara	14	7	14	21	7.0%	0.740
Kolar	15	6	15	22	8.0%	0.740
Belagavi	16	7	15	22	5.8%	0.740
Yadgir	17	5	16	24	10.4%	0.740
Koppal	18	7	16	24	6.8%	0.740

Interpretation: a narrow rank band and high top-5 frequency indicate a stable prioritisation signal. A wide band indicates that KSISU should not over-interpret the ranking until the district evidence improves.

Top-5 frequency across shared Monte Carlo draws (%)



8. Validation, provenance and reproducibility

Test	Result	Interpretation
District linkage	31 / 31 matched	canonical LGD mapping
Seeded typo recovery	80 / 80 (100.0%)	correctness, not just non-empty match
Unseen aliases	0 matched; 20 review	unknown labels are not treated as successes
Duplicates	5 injected; 5 removed	real dedupe pass
Rows dropped	0	unmatched evidence stays reviewable
Monte Carlo	5,000 draws; seed 20261016	reproducible
Statewide conservation	0.935 median/expected	catches allocation errors

Each source artifact carries file_id, SHA-256, URL, year, grain and locator. The run audit log records seed, draw count, source files and output files.

9. What KSISU should deploy next

Phase	Action	Decision enabled
1. Evidence registry	Preserve raw artifacts, source IDs, hashes and LGD/NIC/NCO mapping	reproducible evidence layer
2. Full NCVT district supply	Load complete district x trade ITI export	replace proxy supply with district actual
3. PLFS occupation x industry	Extract state occupation mix from microdata	replace analyst occupation prior
4. District manufacturing employment	Use DSDP/MIS or stronger district industry evidence	reduce allocation uncertainty
5. Production	Scheduled refresh, quality flags, human review, machine-readable outputs	continuous KSISU intelligence

Submission position: SCI, VOI, confidence and evidence gates are proposed methodology for the Skillathon, not the official Karnataka Skill Index formula.

Current blocker: the complete current NCVT MIS district x trade seat export must be loaded before a district-wide policy call can be made. The bundled partial rows are deliberately not presented as complete.

AI-use disclosure: AI assistance was used for code drafting, documentation and analysis design. Public-source provenance, assumptions and evidence tiers are retained in the package.

Sources: LGD Download Directory; Government Open Data Udyam district employment resource; MoSPI ASI 2022-23; NCVT MIS ITI Count / ITI Search / FileDownload; DGT/DITE tracer study; PLFS official portal.